Data Flow Diagram

Wandr AI – AI Travel Itinerary Planner

Level 0 – Context Diagram

External Entities	User (Browser), Supabase Auth, Groq AI API
Central Process	Wandr AI System
Primary Data Flows	Trip requests IN; Itineraries OUT; Auth tokens IN/OUT

Level 1 – Major Processes

Process ID	Process Name	Input	Output	Data Store
P1	Authenticate User	Email + Password	JWT Access Token	Supabase Auth
P2	Generate Itinerary	GenerateTripRequest (JSON)	Trip object (JSON)	D1: trips table
P3	Store Trip	Parsed ItineraryDay[] + metadata	UUID trip ID	D1: trips table
P4	Retrieve Trips	user_id + page params	Paginated TripSummary[]	D1: trips table
P5	Regenerate Day	RegenerateDayRequest + trip context	Updated ItineraryDay	D1: trips table
P6	Calculate Budget	days_json from D1	BudgetBreakdown JSON	In-memory (budget_service.py)
P7	Verify JWT	Authorization: Bearer header	Decoded user payload	Supabase JWKS endpoint

Data Stores

ID	Name	Technology	Contents
D1	trips	Supabase PostgreSQL	id, user_id, title, destination, dates, budget, days_json (JSONB), budget_breakdown (JSONB), timestamps

D2	users	Supabase Auth	id, email, created_at (managed by Supabase)
D3	AI Cache	None (stateless)	Each Groq call is stateless; no caching in v1

Key Data Flows – Description

1. User submits signup → Supabase Auth creates user → Returns JWT.
2. Frontend attaches JWT as Bearer token to every API request.
3. FastAPI verifies JWT via Supabase JWKS (RS256) before processing.
4. Trip generation request hits ai_service.py → Groq API called → JSON parsed → stored in Supabase.
5. RLS policy ensures SELECT/INSERT/UPDATE/DELETE only for auth.uid() = user_id.